29/11/2025

Code-A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 NBTS pre **Time : 44 Min.**

physics -12

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phy-11

Chemistry

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chem-11

BOTANY -12

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BOT-11

ZOOLOGY-12

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181. (3)

ZOO-11

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Hints and Solutions

physics -12

(1) **Answer :** (1)
Solution:

$Y = \overline{\overline{A+B}} = A+B$

(2) **Answer :** (1)
Solution:

Binding energy per nucleon = $\frac{\text{Binding energy of nucleus}}{\text{No. of nucleons}} = \frac{\Delta mc^2}{A}$

(3) **Answer :** (2)
Solution:
Stimulated emitted radiation in the microwave form is called MASER.

(4) **Answer :** (2)
Solution:

$\mathrm{E} = \frac{10^{-3}\times(3\times10^8)^2}{36\times10^5} = 2.5\times10^7 \text{ kWh}$

(5) **Answer :** (3)

(6) **Answer :** (1)
Solution:

$\because n_e n_h = n^2$

$2.25\times10^{22}\times n_h = (1.5\times10^{16})^2$

$2.25\times10^{22}\times n_h = 2.25\times10^{32}$

$n_h = 10^{10}\ \text{m}^{-3}$

(7) **Answer :** (2)
Hint:

$r \propto A^{1/3}$

Solution:

$\frac{r_{Te}}{r_{Al}} = \left(\frac{125}{27}\right)^{1/3} = \frac{5}{3}$

$\therefore\ r_{Te} = 3.6\times\frac{5}{3} = 6\,\text{fm}$

(8) **Answer :** (1)
Hint:

Radius of nuclei $R = R_0 A^{\frac{1}{3}}$

Solution:

Radius of nucleus of ${}_4\text{Be}^9 = R$

Radius of ${}_{32}X$ nucleus $= 2R$

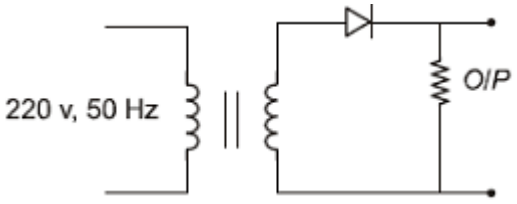
$\frac{R}{2R} = \left(\frac{9}{A}\right)^{\frac{1}{3}}$

$\left(\frac{1}{2}\right)^3 = \frac{9}{A}$

$A = 72$

Number of neutrons in nucleus $X = 72 - 32 = 40$

(9) **Answer :** (3)
Hint:
Half wave rectifier output current.
Solution:



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v_m for diode

$v_m = \frac{210\sqrt{2}}{10} = 21\sqrt{2}V$

As $I_m = \frac{v_m}{R_F + R_L} = \frac{21\sqrt{2}}{70}$

$= \frac{3\sqrt{2}}{10}$

$= \frac{3\sqrt{2}}{5\times 2} = \frac{3}{5\sqrt{2}}A$

(10) **Answer :** (2)
Solution:

$\frac{1}{\lambda} = RZ^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$

$\frac{1}{\lambda'} = RZ^2\left[\frac{1}{2^2} - \frac{1}{4^2}\right]$

$\therefore\ \lambda' = \frac{20\lambda}{27}$

(11) **Answer :** (1)
Solution:
Light emitting diodes (LED) are used in forward bias for their operation.

(12) **Answer :** (2)
Solution:
Reverse bias applied to a junction diode increases the potential barrier.

(13) **Answer :** (1)
Solution:

$v_n = \frac{Zv_0}{n}$

$\frac{v_4}{v_2} = \frac{2}{4}$

$\Rightarrow \quad v_4 = \frac{v_2}{2} = \frac{v}{2}$

(14) **Answer :** (2)
Solution:
Isotones have same number of neutrons but different number of protons.

(15) **Answer :** (4)

(16) **Answer :** (1)

(17) **Answer :** (3)
Solution:
In the reaction sum of atomic number remains the same but mass reduces by 3.
Hence 3 neutrons were produced.

(18) **Answer :** (2)
Solution:
N(θ) = $\frac{k}{\sin^4(\theta/2)}$

Hence *N*(θ) decreases with increase in angle θ.

(19) **Answer :** (1)

(20) **Answer :** (1)
Solution:
In hydrogen atom

$r = \frac{r_0 \times n^2}{Z} \ \Rightarrow\ r \propto n^2$

$v = \frac{v_0 \times Z}{n} \Rightarrow\ v \propto \frac{1}{n}$

$T = \frac{2\pi r}{v} \Rightarrow\ T \propto \frac{\frac{n^2}{z}}{\frac{z}{n}} \Rightarrow\ T \propto n^3$

$B = \frac{\mu_0 \mathrm{i}}{2r} = \frac{\mu_0 e}{2rT} \quad \Rightarrow B \propto \frac{1}{n^2 \times n^3} \ \Rightarrow B \propto \frac{1}{n^5}$

(21) **Answer :** (2)

(22) **Answer :** (2)
Solution:

$\frac{1}{\lambda_L} = R\left[\frac{1}{1^2} - \frac{1}{\infty}\right] = R$

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$\frac{1}{\lambda_B} = R\left[\frac{1}{2^2} - \frac{1}{\infty}\right] = \frac{R}{4}$

$\frac{1}{\lambda_p} = R\left[\frac{1}{3^2} - \frac{1}{\infty}\right] = \frac{R}{9}$

$\lambda_L : \lambda_B : \lambda_P = \frac{1}{R} : \frac{4}{R} : \frac{9}{R} = 1 : 4 : 9$

(23) **Answer :** (1)
Solution:
Fusion process requires very high temperature to overcome coulomb potential barrier. Electron and positron are particle-antiparticle pair.

(24) **Answer :** (3)

(25) **Answer :** (2)
Solution:
ΔE = 7(4) – [2(1.1)] × 2 = 28 – 4.4 = 23.6 meV

(26) **Answer :** (3)

(27) **Answer :** (3)
Solution:
Apply momentum conservation
$0 = Mv + \frac{E}{c}$
$v = \frac{E}{Mc}$
Recoil energy of nucleus = $\frac{1}{2}Mv^2 = \frac{E^2}{2Mc^2}$

(28) **Answer :** (2)
Solution:
The moderator slows down neutron by decreasing its KE through a series of elastic collisions

(29) **Answer :** (3)

(30) **Answer :** (1)
Hint:
It is a balanced Wheatstone bridge.
Solution:
$R_{\text{eff}} = \frac{R_1R_2}{R_1+R_2} = \frac{6\times12}{6+12} = 4\,\Omega$

(31) **Answer :** (2)
Solution:
Velocity = Slope (x – t) graph
At B, the slope is zero.

(32) **Answer :** (2)
Solution:
Average velocity is a vector quantity. Its magnitude is leas than or equal to average speed.

(33) **Answer :** (1)
Hint:
Use equation of uniformly accelerated motion.
Solution:
$2as = v^2 - u^2$
$2 \times a \times 6 = \left(\frac{u}{2}\right)^2 - u^2$ ...(i)
$2 \times a \times d = 0 - \left(\frac{u}{2}\right)^2$ ...(ii)
By dividing equation (i) and (ii), we get

$$\frac{2\times a\times 6}{2\times a\times d} = \frac{\frac{-3u^2}{4}}{-\frac{u^2}{4}}$$

$\frac{6}{d} = 3$
$\Rightarrow d$ = 2 cm

(34) **Answer :** (2)
Hint:
Use $\frac{1}{v} = \frac{dt}{dx}$
Solution:
$t = 3x^2 + 2x$
$\frac{dt}{dx} = 6x + 2$

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$v = \frac{1}{6x+2}$

$a = \frac{dv}{dt} = \frac{d}{dx}\left(\frac{1}{6x+2}\right).\frac{dx}{dt}$

$= \frac{-6}{(6x+2)^2}\left(\frac{1}{(6x+2)}\right)$

$= \frac{-6}{(6x+t)^3}$

$a = -6v^3$

(35) Answer : (1)
Hint:

Average speed = $\frac{\text{Total distance}}{\text{Toal time}}$

Solution:

Average speed = $\frac{2v_1v_2}{v_1+v_2} = \frac{2\times20\times60}{20+60} = \frac{2\times20\times60}{80}$ = 30 km/h

(36) Answer : (1)
Solution:

$a = \frac{\Delta v}{\Delta t}$

$\Delta v = a\Delta t$

$v_f - 0 = 5 \times 10$

v = 50 m/s

(37) Answer : (1)
Solution:
When acceleration is in opposite to velocity, speed will decrease irrespective of acceleration is increasing or decreasing only rate may be more or less.

(38) Answer : (4)

(39) Answer : (1)
Solution:

$a = v \cdot \frac{dv}{ds}$

$\frac{dv}{ds} = \frac{25}{100} = \frac{1}{4}\ s^{-1}$

$\frac{v-50}{50} = \frac{1}{4}\ s^{-1}$

v = 62.5 m/s

$a = 62.5 \times \frac{1}{4} = \frac{125}{8}$ m/s^2

(40) Answer : (3)
Hint:

Use, $v = \left|\frac{dx}{dt}\right|$

Solution:

$x = 5 - 4t + t^2$

$v = \frac{dx}{dt} = -4 + 2t$

For speed

$v = \left|\frac{dx}{dt}\right| = \left|-4 + 2t\right|$

(41) Answer : (4)
Solution:
The distance covered will always be equal or greater than magnitude of displacement covered by body. Hence, ratio $\frac{\text{distance}}{\text{displacement}} \geq 1$.

(42) Answer : (2)
Hint:

Use, $a = \frac{v-u}{t}$

Solution:

$t = \frac{20-0}{2}$

= 10 s

$S = v_{avg} \times t$

= 20 × 10

= 200 m

(43) Answer : (4)
Solution:

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$$\frac{S_5}{S_{5^{th}}} = \frac{\frac{1}{2}g\left(5\right)^2}{\frac{1}{2}g\left[2\times5-1\right]} = \frac{25}{9}$$

(44) Answer : (2)
Solution:

To reach water $t_1 = \sqrt{\frac{2h}{g}}$

Now sound of splash will come back with velocity of sound $t_2 = \frac{h}{v}$

$t = t_1 + t_2 = \sqrt{\frac{2h}{g}} + \frac{h}{v}$

(45) Answer : (4)
Solution:

$T_{\text{still water}} = \frac{2\,t_u t_d}{t_u + t_d}$

phy-11

Chemistry

(46) Answer : (4)

(47) Answer : (4)
Hint:
H_2SO_4 gives charring action with carbohydrates
Solution:

- $\underset{\text{(A)}}{\mathrm{Ca_3\,P_2}} + 6\mathrm{H_2O} \rightarrow 2\,\mathrm{PH_3\,(g)} + 3\,\mathrm{Ca\,(OH)_2}$
- $2\,\mathrm{PH_3}\left(\mathrm{g}\right) + \mathrm{CuSO_4} \rightarrow \underset{\text{(B)}}{\mathrm{Cu_3\,P_2}} + \underset{\text{(C)}}{3\mathrm{H_2\,SO_4}}$
- PH_3 is a colorless gas with rotten fish smell. It is slightly soluble in water. It explodes in contact of trace amount of HNO_3.

(48) Answer : (3)
Solution:

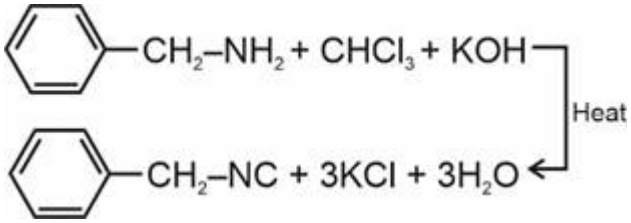
a. $2\,\mathrm{SO_2} + \mathrm{O_2} \xrightarrow{\mathrm{V_2O_5}} 2\,\mathrm{SO_3}$

b. $4\mathrm{HCl} + \mathrm{O_2} \xrightarrow{\mathrm{CuCl_2}} 2\,\mathrm{Cl_2} + 2\mathrm{H_2O}$

c. $4\,\mathrm{NH_3} + 5\mathrm{O_2} \xrightarrow[500\mathrm{K}/9\text{ bar}]{\mathrm{Pt\,/\,Rh\ gauge}} 4\mathrm{NO} + 6\mathrm{H_2O}$

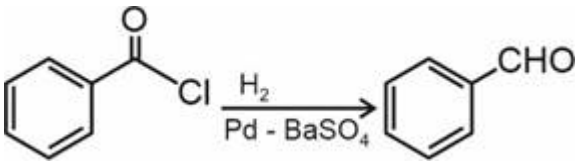
d. $\mathrm{N_2} + 3\mathrm{H_2} \xrightarrow[773\mathrm{K}]{\mathrm{Fe}} 2\,\mathrm{NH_3}$

(49) Answer : (1)
Hint:
Aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide solution form isocyanides or carbylamines.
Solution:



Above reaction is known as carbylamine reaction.

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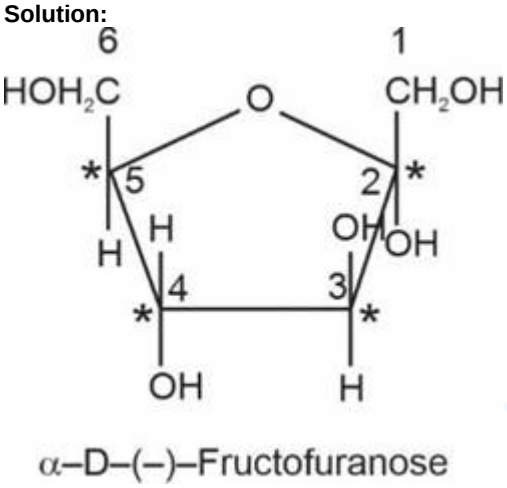


Reaction is known as Rosenmund reaction.

(50) **Answer :** (4)
Solution:

Name of amine	pK_b
Methanamine	3.38
N-Methylmethanamine	3.27
N, N-Dimethylmethanamine	4.22
Ethanamine	3.29

(51) **Answer :** (2)
Hint:
Carbon with all four different valencies is chiral carbon.
Solution:



α–D–(–)–Fructofuranose

• (*) represents chiral carbon

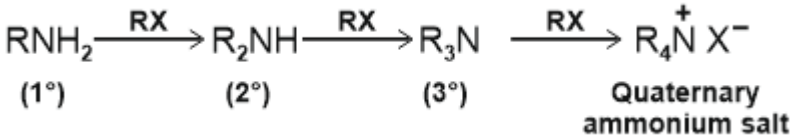
(52) **Answer :** (2)
Solution:

$$CuSO_4 + H_2S \rightarrow \underset{\text{Black}}{CuS} + H_2SO_4$$

$$CuS + 2HNO_3 \rightarrow \underset{\text{Blue solution}}{Cu(NO_3)_2} + H_2S$$

$$Cu(NO_3)_2 + 4NH_3 \rightarrow \underset{\text{Deep blue solution}}{[Cu(NH_3)_4]^{2+}} + 2NO_3^-$$

(53) **Answer :** (4)
Solution:

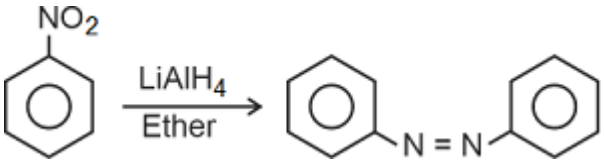


(54) **Answer :** (3)
Solution:

Compound	Boiling point
NH_3	238.5 K
PH_3	185.5 K
AsH_3	210.6 K
SbH_3	254.6 K

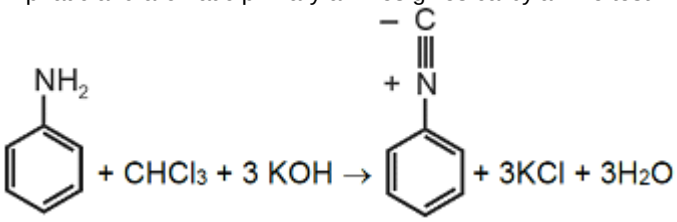
(55) **Answer :** (3)
Solution:

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(56) **Answer :** (4)
Solution:
Size of colloidal particle is from 1-1000 nm.
Starch sol is lyophilic and stability of lyophobic sols is due to charge on colloidal particles.

(57) **Answer :** (3)
Solution:
Aliphatic and aromatic primary amines gives carbylamine test.



(58) **Answer :** (1)
Hint:
Tertiary amine does not react with Hinsberg's reagent ($PhSO_2Cl$)

(59) **Answer :** (1)
Solution:

Compound	Boiling point (K)
NH_3	238.5
PH_3	185.5
AsH_3	210.6
SbH_3	254.6
BiH_3	290

(60) **Answer :** (1)
Hint:
Tertiary amine does not react with Hinsberg's reagent ($PhSO_2Cl$)

(61) **Answer :** (1)
Solution:
Amino acids combines to form proteins.

(62) **Answer :** (3)
Solution:
Starch forms iodo starch complex with Iodine which is blue coloured.

(63) **Answer :** (4)
Hint:
Partial hydrolysis of XeF_6 gives different products.
Solution:

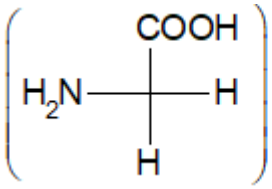
$$XeF_6 + 2H_2O \rightarrow \underset{(x)}{XeO_2F_2} + 4HF$$

Shape of X is see-saw and hybridisation of xenon in XeO_2F_2 is sp^3d

(64) **Answer :** (3)
Solution:
As the oxidation state of Halogen increases acidic strength of oxyacid increases.
Therefore correct order of acidic strength is
$HClO < HClO_2 < HClO_3 < HClO_4$

(65) **Answer :** (3)
Hint:

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Glycine  is non-essential amino acid.

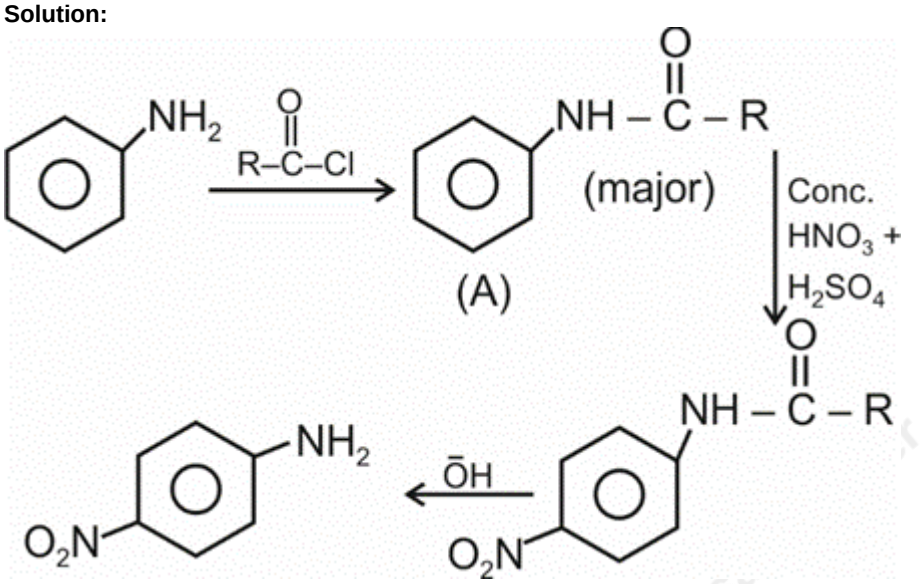
(66) **Answer :** (2)
Solution:
Convulsions is caused by the deficiency of vitamin B_6
Scurvy is caused by the deficiency of vitamin C.

(67) **Answer :** (1)
Solution:
Vitamin-D causes rickets.

(68) **Answer :** (1)
Solution:

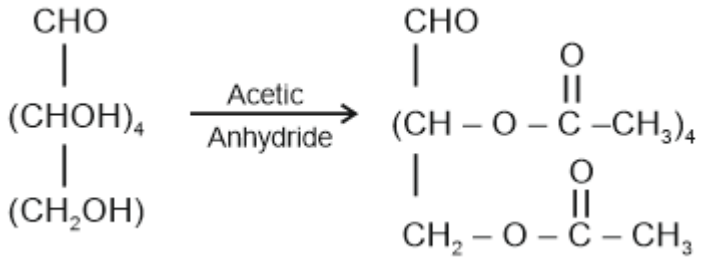
$$H_3C-N=C=O \xrightarrow[\text{Hydrolysis}]{\text{Alkaline}} CH_3-NH_2$$

(69) **Answer :** (2)
Solution:



(70) **Answer :** (4)
Solution:
PbS – Black ppt
$PbCrO_4$ – Yellow ppt
AgBr – Pale yellow ppt
$PbSO_4$ – White ppt

(71) **Answer :** (2)
Solution:



(72) **Answer :** (1)
Solution:
$(NH_4)_2[Ce(NO_3)_6] + 3ROH \rightarrow [Ce(NO_3)_4(ROH)_3] + 2NH_4NO_3$

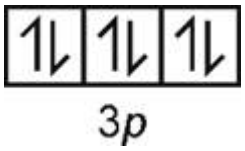
(73) **Answer :** (1)

(74) **Answer :** (3)

Solution:
The correct order is Cl > F > Br > I

(75) **Answer :** (3)
Solution:
Albumin is a globular protein.
Globular proteins are usually soluble in water.

(76) **Answer :** (4)
Hint:



Only three electrons have $+\frac{1}{2}$ and other three have $-\frac{1}{2}$ spin quantum number.
Solution:

	Number of electrons
K^+	18
Ca^{2+}	18
S^{2-}	18

These are isoelectronic species
• Shape of orbital is determined by Azimuthal quantum number.

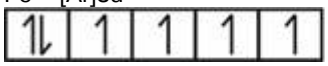
(77) **Answer :** (1)
Solution:
$E_n = -13.6\frac{Z^2}{n^2}$ = –1.51 eV
$n^2 = \frac{13.6}{1.51} = 9$
n = 3
$r_n = 0.529 \times \frac{n^2}{Z}$ Å = 0.529 × $\frac{9}{1}$ Å = 4.76 Å

(78) **Answer :** (2)
Solution:
Mass of 1 gram atom $= \frac{25}{0.25} = 100$
∴ Atomic mass = 100

(79) **Answer :** (4)
Solution:
H_2 is limiting reactant
30 ml of H_2 will react with 15 ml of O_2
as $2H_2 + O_2 \rightarrow 2H_2O$
So V_{o_2} left unreacted = 20 – 15 = 5 ml

(80) **Answer :** (4)
Solution:
$r_n = r_0 \times \frac{n^2}{Z}$
Radius of second orbit of $He^+ = r_0 \times \frac{2\times2}{2} = 2\,r_0$
r_0 is the radius of 1st orbit of hydrogen atom.
Radius r_0 is proportional to n^2

(81) **Answer :** (1)
Hint:
Electronic configuration of Fe = $[Ar]3d^64s^2$
Solution:
Fe = $[Ar]3d^6$



(82) **Answer :** (3)
Hint:
In case of addition, the final result should be reported based on the number carrying minimum number of decimal places to the right
Solution:

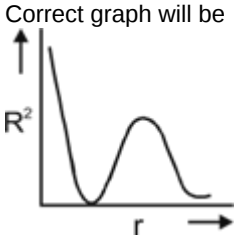
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3.26 + 0.023 = 3.28

(83) **Answer :** (4)
Solution:
When electron moves in an orbit neither loose nor gain energy

(84) **Answer :** (3)
Hint:
n = 4 and *l* = 2 is 4*d*
Solution:
d orbitals are five in number so one *d* orbital will occupy maximum 2 electrons.

(85) **Answer :** (1)
Solution:
Graph is not correct [because 2s have only one node]
Correct graph will be



(86) **Answer :** (4)
Solution:
Solutions are isotonic π $p_{urea} = p_{unknown}$
$\Rightarrow C_{urea} = C_{unknown}$ (at same T)

$\Rightarrow \quad \frac{\frac{20}{60}\ \text{mol}}{1\,\text{L}} = \frac{\frac{5}{x}\ \text{mol}}{\frac{100}{1000}\,\text{L}} \left(\because\ 5\% = \frac{5\,\text{g}}{100\ \text{mL}}\right)$

$\Rightarrow \quad \frac{2}{6} = \frac{50}{x}$

$\Rightarrow$ x = 150 g mol^{-1}

(87) **Answer :** (2)
Hint:
Number of electrons in one molecule of water = 10
Solution:
Number of electrons $= \frac{36}{18} \times 6 \times 10^{23} \times 10$
= 12 × 10^{24}

(88) **Answer :** (1)
Solution:

Molarity of NH_3 solution = $\frac{18.6 \times 0.93 \times 10}{17}$ = 10.17 M
Strength (g/L) = molarity × mol. mass
= 10.17 × 17 = 172.9 g/L
i.e., 1000 mL of solution contain NH_3 = 172.9 g
1 mL or 1 cm^3 of solution contain NH_3 = $\frac{172.9}{1000}$ = 0.172 g ≈ 0.17 g

(89) **Answer :** (4)
Solution:

$63.546 = \frac{63(x)+65(100-x)}{100}$
$\Rightarrow$ x = 72.7

(90) **Answer :** (1)
Solution:
Electronic configuration of Cu^+ : [Ar] $3d^{10}$

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chem-11

BOTANY -12

(91) **Answer :** (2)
Solution:
Habitat loss and fragmentation is the mort important cauge driving animals and plants to extinction because when large habitats are broken up into small fragments due to various human activities, mammals and birds requiring large territories and certain animals with migratory habits are badly affected, leading to population declines.

(92) **Answer :** (4)

(93) **Answer :** (2)
Solution:
Major site of decomposition in the soil and water bodies occur on upper and bottom layer respectively.

(94) **Answer :** (3)
Solution:
The extinction of passenger pigeon was due to over exploitation by humans.

(95) **Answer :** (3)
Solution:
The regression coefficient (z) values in the range of 0.6 to 1.2 for very large areas.

(96) **Answer :** (1)
Solution:
India has only 2.4% of world's land area but it possesses 8.1% species diversity of the world.

(97) **Answer :** (4)
Solution:
Lantana and African catfish are exotic species of India.
Nile perch is exotic species of East Africa.

(98) **Answer :** (3)
Solution:
Passenger pigeon became extinct mainly due to overexploitation by humans.

(99) **Answer :** (1)
Solution:
The standing crop is measured as the mass of living organismor the number in a unit area. It can be defined as amount of living material present in different trophic levels at a given time.

(100) **Answer :** (2)
Solution:
In a population, a species susceptible to extinction does not have the ability to switch over to alternate food source.

(101) **Answer :** (1)
Hint:
Biomass is the amount of living matter, expressed as dry weight in kg / m^2.
Solution:
If total biomass of producers in a specific area is more than that of primary consumers and gradually decreases at each successive trophic level, forms upright pyramid.

(102) **Answer :** (2)
Solution:
Endemism means species confined to that region and not found anywhere else.

(103) **Answer :** (4)
Solution:
The amount of energy needed to sustain the pyramid enters at level A. The total amount of energy decreases with each successive feeding level from A to D. The amount of energy is different in each level of the pyramid, as energy flow follows 10% energy law.

(104) **Answer :** (1)
Solution:
In global biodiversity of plants, Fungi has maximum diversity followed by angiosperms.

(105) **Answer :** (3)
Solution:

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Rate of decomposition will be high when moisture and aeration are optimum.

(106) Answer : (3)
Solution:
Upright pyramids indicate that given parameters at lower trophic level are more than at a higher level.
In most ecosystems all upright pyramids of energy, number and biomass indicates that producers are more in number and biomass than the herbivores and energy at a lower trophic level is always more than at a higher level.

(107) Answer : (3)
Solution:
Food web is more realistic form of transfer of energy & food in nature than food chain.
The natural interconnection of food chains make food web, which provides stability to ecosystem. It operates because of taste preference and unavailability of food.

(108) Answer : (4)
Solution:
The largest man made ecosystem is agroecosystem.

(109) Answer : (4)
Solution:
According to 10% law of energy transfer, sparrow has much energy as compared to other prey organisms for hawk.

(110) Answer : (3)
Hint:
Hotspots are areas of high endemism and very high level of species richness.
Solution:
Hotspots are most threatened reservoir of plants and animals life on earth.

(111) Answer : (3)
Solution:
Stratification is a structural component of ecosystem.

(112) Answer : (2)
Solution:
Species diversity is diversity of species within a region.

(113) Answer : (1)
Solution:
Cow and Grasshopper are primary consumers.

(114) Answer : (4)
Solution:
The fast dwindling Amazon forest is estimated to produce 20% of the total oxygen in Earth's atmosphere.

(115) Answer : (3)
Solution:
Biodiversity loss occurs when biological systems are exploited beyond the degree of their renewability.

(116) Answer : (1)
Solution:
Grazing food chain is the major conduit of energy flow in aquatic ecosystems.

(117) Answer : (1)
Solution:
The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceeds that of phytoplanktons.

(118) Answer : (2)
Hint:
The growth of multicellular organisms is due to mitosis.
Solution:
Meiosis increases the genetic variability in the population of organisms from one generation to the next.

(119) Answer : (3)
Solution:
Mitochondria is the site of aerobic respiration.
Both mitochondria and chloroplasts

1. Are double membrane bound.
2. Have dsDNA
3. Execute ATP synthesis
4. Have ETS.

But only mitochondria is the site for oxidation of respiratory substrates like carbohydrates, proteins, fats, etc.

(120) Answer : (4)
Solution:

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G_1 to S transition is carried out by G_1 cyclin + cdc 2 kinase.
Cyclins regulate eukaryotic cell cycle.

(121) Answer : (3)
Solution:
During meiosis I and mitosis nuclear membrane disappears and microtubules attach to the kinetochores of chromosomes.

(122) Answer : (2)
Solution:
Final stage of prophase I is marked by Diakinesis. Diplotene stage can last for months or years in ooctyes of vertebrates some.

(123) Answer : (2)
Solution:
Cytokinesis in plant cell is achieved by cell plate formation, which is formed in centrifugal manner.

(124) Answer : (3)
Hint:
Anaphase II is similar to mitotic anaphase.
Solution:
In anaphase I, the homologous chromosomes separate from each other and start moving towards the poles.

(125) Answer : (3)
Hint:
Mitosis is also called equational division.
Solution:
Meiosis helps in increasing the genetic variability in the daughter cells because it involves the crossing over of chromosomes.

(126) Answer : (1)
Hint:
Mesosomes are in the form of vesicles, tubules and lamellae.
Solution:
Mesosomes are involved in cell wall formation, DNA replication and distribution to daughter cells.

(127) Answer : (3)
Solution:
Cell plate formation begins in centre.
In plant cell, cell plate formation begins in centre and it grows towards the periphery. In animal cell, cytokinesis is done with the help of cell furrow method.

(128) Answer : (3)
Hint:
Mitochondrial DNA resembles to bacterial DNA.
Solution:
Mitochondrial DNA is double stranded and circular.

(129) Answer : (3)
Hint:
Chlorophyll traps light energy for photosynthesis.
Solution:
Ribosomes are the site of protein synthesis. Chromoplasts contain carotenoid pigments and amyloplast stores starch.

(130) Answer : (2)
Solution:
Double metaphasic plate is formed during metaphase I of meiosis-I

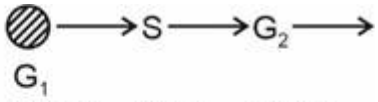
(131) Answer : (3)

(132) Answer : (2)
Solution:
Sole members of the kingdom Monera are bacteria and they lack membrane-bound organelles. Thus, they do not have mitochondria, chloroplast and peroxisome.

…[1 Mark]

(133) Answer : (1)
Hint:
In S phase – DNA content doubles but chromosome number remains the same.
Solution:

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	G_1	S	G_2
DNA content	30 pg	60 pg	60 pg
Chromosome no.	20	20	20

(134) Answer : (2)

Hint:

Special membranous structure is the mesosome which is formed by the extensions of plasma membrane into the cell.

Solution:

In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

(135) Answer : (3)

Hint:

Decomposition is largely an oxygen requiring process.

Solution:

Decomposition rate is quicker if detritus is rich in nitrogen and water-soluble substance like sugars.

BOT-11

ZOOLOGY-12

(136) Answer : (2)

Solution:

In 1983, Eli lily an American company prepared two DNA sequences corresponding to A and B chains of insulin and introduced them in plasmids of *E.coli*. Chain A and B were produced separately, extracted and combined by creating disulphide bonds to form human insulin.

(137) Answer : (2)

Hint:

Organism which produces streptokinase.

Solution:

Streptokinase produced by *Streptococcus* bacterium and modified by genetic engineering is used as a 'Clot buster' for removing clots from blood vessels of patients who have undergone myocardial infarction.

(138) Answer : (1)

Solution:

Probe is a single stranded DNA or RNA, tagged with a radioactive molecule which is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.

(139) Answer : (1)

Solution:

NCERT Reference: Page= 196, Zoology_12th

(140) Answer : (1)

Solution:

Ori site is a sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA in the vector. So, if one wants to recover many copies of the target DNA, it should be cloned in a vector whose origin support high copy number.

(141) Answer : (1)

Solution:

DNA profile of suspect number 3 is completely matching with the DNA from the blood from the crime scene.

(142) Answer : (1)

Hint:

Condition that creates transient pores in cell membrane.

Solution:

42°C creates pores in lipid membrane allowing entry of DNA. Cells can be made competent by treating them with ice cold $CaCl_2$.

(143) Answer : (2)

Solution:

Smaller DNA fragments move faster towards anode while larger fragments remain closer to the well. Ethidium bromide is used to visualise DNA under UV light. Extraction of DNA from the gel is known as elution.

(144) Answer : (1)
Solution:
Strains of *Bacillus thuringiensis* are used to produce bio insecticidal plants.

(145) Answer : (3)
Hint:
DNA ligase
Solution:
DNA ligases also known as molecular glue belong to class VI of enzymes, catalyse the linking together of alien DNA and cloning vector.
Restriction endonucleases also known as molecular scissors help in cutting the DNA at specific locations.

(146) Answer : (1)
Solution:
The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.

(147) Answer : (1)
Hint:
It is produced by *Streptococcus*.
Solution:
Myocardial infarction can now be treated by genetic engineering. Erythropoietin is synthesized from J.G. cells of kidney promoting erythropoiesis. Interleukins are lymphokines that activate immune system.

(148) Answer : (2)
Hint:
It is genetic material of *Reoviruses*
Solution:
In nature, RNA is used for the regulation of specific genes and is also applied as a defence against pathogens.

(149) Answer : (1)
Solution:
Serum and urine analysis are conventional methods of disease diagnosis which takes time. Widal test is a diagnostic test for typhoid.

(150) Answer : (2)
Solution:
siRNA binds with specific mRNA of nematode and prevents its translation.

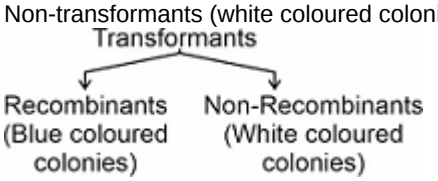
(151) Answer : (2)
Solution:
Insulin, used for treating diabetes mellitus was earlier extracted from pancreas of slaughtered cattle and pigs, but its administration caused allergy to some patients.

(152) Answer : (1)
Solution:
Alkaline pH of midgut activates the protoxin

(153) Answer : (3)
Solution:
Two of the major cotton pests are the boll weevil and the pink bollworm.

(154) Answer : (2)
Solution:
Contact inhibition, meiotic divison and presence of inactivated oncogenes are character of normal cells.

(155) Answer : (3)
Solution:
The Lac-Z gene codes for the β-galactosidase. The enzyme utilises its substrate to produce blue coloured product.
In the given case,
Non-transformants (white coloured colonies)



(156) Answer : (4)
Hint:
Identify a super bug.
Solution:

Bioremediation is a strategy that uses naturally occurring organisms to breakdown pollutants, such as petroleum hydrocarbons into less toxic substances. Bioremediation by *Pseudomonas aeruginosa* is also well documented.

(157) Answer : (4)
Solution:
Neem: In 1996, Vandana Shiva challenged the patent granted to the firm of W.R. Grace & Co. by the European Patent Office, Munich for 'fungicidal uses of neem oil'.
Turmeric: In May, 1995 the US Patent Office granted to the university of Mississippi Medical Center a patent for "Use of Turmeric in Wound Healing".
Daffodil: It is favourite plant for taking genes for synthesis of β-carotene.

(158) Answer : (4)
Solution:
Retrovirus mediated vector is used as an indirect method of gene transfer.

(159) Answer : (3)
Solution:
Rosie is the first transgenic cow that produced human protein enriched milk containing α-lactalbumin.

(160) Answer : (3)

(161) Answer : (3)
Solution:
DNA fragments being negatively charged move towards anode and get resolved according to their size.

(162) Answer : (1)
Solution:
Genetically modified crops are used to minimise the use of fertilizers and chemicals so that their harmful effects on the environment are reduced. Genetic modification has been used to create tailor made plants.

(163) Answer : (2)
Solution:
Restriction site for *Bam*H I and *Sal* I are present in the tetracycline resistance gene.

(164) Answer : (4)
Solution:
Genetically modified plants have ability to increase efficiency of mineral usage of plants. It prevents early exhaustion of fertility of soil.

(165) Answer : (3)
Solution:
Selectable marker helps in identifying and eliminating non-transformants and selectively permitting the growth of transformants.

(166) Answer : (1)

(167) Answer : (1)
Solution:
Cuboidal epithelium constitutes cube-like cells whereas columnar epithelium is made up of tall cells.

(168) Answer : (4)
Solution:
Connective tissue is the most abundant and widely distributed tissue in the body of complex animals. It includes bone, cartilage, adipose tissue, aerolar tissue, blood, bone and adipose tissue.

(169) Answer : (1)
Hint:
Muscle fibres associated with limbs and heart are striped in appearance.
Solution:
Striated muscles include both cardiac & skeletal muscles. The alternate light and dark bands refer to the horizontal arrangement of actin & myosin filaments.

(170) Answer : (3)
Hint:
They perform cementing between cells
Solution:
Adhering junctions are also called macula adherens. They possess intercellular proteins and tonofibrils which keep the neighbouring cells together.

(171) Answer : (1)

(172) Answer : (2)
Solution:
Epithelial tissue has
– Little intercellular matirx
– Does not cover all part of body

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(173) Answer : (2)
Solution:
Compound epithelium is made up of more than one layer (multi layers) of cells and thus has a limited role in secretion and absorption. They cover the dry surface of skin, the moist surface of buccal cavity, pharynx etc.

(174) Answer : (3)
Solution:
Compound epithelium provides protection against mechanical and chemical stresses.

(175) Answer : (4)
Solution:
Epithelial tissues usually lies on the non-cellular basement membrane which consists of two layers i.e. basal and fibrous lamina.

(176) Answer : (2)
Hint:
Neuroglial cells protect and support neurons.
Solution:
Neuroglial cells are:
· Oligodendrocytes and Schwann cells – Secrete myelin sheath
· Astrocytes – Form blood brain barriers
· Ependymal cells – Lining of ventricles and central canal.

(177) Answer : (2)
Solution:
Elastic cartilage is present at the tip of nose.

(178) Answer : (4)
Hint:
Similar location as that of adipose tissue
Solution:
Dense irregular connective tissue is present in the skin.
• Tendons attach skeletal muscles to bones.
• Loose connective tissues contain fibroblasts, macrophages and mast cells.
• Bones contain osteocytes.
• Cells present in connective tissues secrete polysaccharides which accumulate between cells and form matrix.

(179) Answer : (2)
Solution:
Blood does not contain fibres.

(180) Answer : (3)
Hint:
Also found in the walls of blood vessels
Solution:
Ciliated epithelium lines the inner surface of fallopian tubes, bronchioles and small bronchi.
Squamous epithelium lines alveoli.

(181) Answer : (3)
Solution:
Recombinant transformants are the ones in which gene of interest is ligated at *Pvu* I recognition site and because site of gene of interest is same as that of recognition site of *Pvu* I, ampicillin resistance would be lost, therefore these colonies will die in medium containing antibiotic ampicillin.
Therefore; recombinant transformants would grow in medium containing tetracycline but not in ampicillin and they do show blue colour in presence of chromogenic substrate. Non-recombinant transformants have taken up the plasmid but not the one which carries the gene of interest; therefore, they grow in medium containing ampicillin and tetracycline antibiotics as they have both amp^r and tet^r genes. Hence, they do not show blue colour in presence of chromogenic substrate.

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